

# Empirical Proof Record: VALIDATED

2026-05-22T18:11:53.104513+00:00

## Empirical Proof Record — VALIDATED

**Proof ID:** d5a4ce3a7c13c2da3c49a12950204c2c...

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### 1. Claim

When an identically-recognised probe is re-exposed across a trajectory, it lands on a strictly-deeper (more-scarred) manifold every time:  $\text{memory\_path\_dependence} = (\text{recognised re-exposure transitions with } \Delta \text{ permanent-scars} > 0) / (\text{recognised re-exposure transitions}) \geq 1.0$ , with the whole-run scar count monotonic non-decreasing (a scar cannot be reset). This measures memory where Kimera keeps it — the scars and the S4 manifold — not the content-recognition layer.

- **Operationalisation:** Run 12 stimuli each re-exposed 3x, interleaved ( $\text{gap} = \text{len}(\text{stimuli})$ ), through Kimera's entity target (Takwin) in batch mode so the substrate accumulates across the batch. Per cycle read `vault_stats.total_scars_stored` (canonical permanent scars), `arachne_web_coupling_frobenius` (manifold deformation), and the concept set (recognition). For each same-stimulus consecutive exposure pair: recognised iff  $\text{Jaccard}(\text{concepts}) \geq 0.80$ ; confirmed iff recognised AND  $\Delta \text{ scars} > 0$ .  $\text{memory\_path\_dependence} = \text{confirmed} / \text{recognised}$ . Forced to 0.0 (REFUTED) if the whole-run scar count ever decreases.
- **Threshold:**  $\text{memory\_path\_dependence} \geq 1.0$  proportion
- **H0:** H0:  $\text{memory\_path\_dependence} < 1.0$  — some re-exposure of a recognised probe did NOT land on more permanent scars (the substrate re-derives without accumulating), or a scar was reset; not path-dependent permanent memory
- **H1:** H1:  $\text{memory\_path\_dependence} \geq 1.0$  — every re-exposure of a recognised probe lands on a strictly-deeper manifold and no scar is ever reset; path-dependent permanent memory in the scar/S4 substrate

## 2. Verdict

**VALIDATED** — observed 1.0

memory\_path\_dependence 1.0000 = 20/20 recognised re-exposure transitions landed on strictly more permanent scars; 0 transitions failed recognition (excluded). Permanence: HOLDS (0 scar-count decreases over the run). Scars 1.0→29.0; coupling 1.240565→16.629817; phi |delta| mean 0.0020 ( $\Phi$ -rigid → memory is in the scars/manifold, not  $\Phi$ ); scar-depth vs exposure-ordinal Spearman rho=0.944, p=3.1e-18 (deepens with experience)

## 3. Evidence

| Pillar                        | Statistic              | Value | 95% CI | Library         |
|-------------------------------|------------------------|-------|--------|-----------------|
| 0.flow.memory_path_dependence | memory_path_dependence | 1.0   | —      | ophamin 0.109.0 |

## 4. Pre-registration

- Registered at: 2026-05-22T18:11:52.760330+00:00
- Config hash: 7d353f8d2b982cf4c2ecb24447af42ad...
- Data hash: 548d6568d12e293ce4229c8104f6265f...

## 5. Signature

3a1eb9735f927eaf6383028eb6782a0f1996f7a0524752849a2e36164e1392b4